

Course: B. Tech. Branch : Computer Science & Engineering Semester : III

Subject Code & Name: Computer Architecture & Organization (BTCOC304)

Max Marks: 60

Date:21/08/25

Duration: 3

Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	Marks
Q. 1 Objective type questions. (Compulsory Question)	12
1 To reduce the memory access time we generally make use of _____	1
a. SDRAM's b. Heaps c. Cache's d. Higher capacity RAM's	
2 In CISC architecture most of the complex instructions are stored in _____	1
a. CMOS b. Register c. Transistors d. Diodes	
3 A source program is usually in _____	1
a. Assembly b. Machine level c. High-level d. Natural language	
language language language	
4 The ALU makes use of _____ to store the intermediate results.	1
a. Accumulators b. Registers c. Heap d. Stack	
5 Which registers can interact with the secondary storage?	1
a. MAR b. PC c. IR d. R0	
6 ISP stands for _____	1
a. Instruction b. Information c. Interchange d. Interrupt Service	
Set Processor Standard Standard Procedure	
Processing Protocol	
7 The addressing mode/s, which uses the PC instead of a general purpose register is _____	1
a. Indexed b. Relative c. Direct d. Both Indexed and direct	
8 Which representation is most efficient to perform arithmetic operations on the numbers?	1

	a. Sign-magnitude	b. 1's complement	c. 2's complement	d. None of the mentioned	
9	When we subtract -3 from 2, the answer in 2's complement form is _____				1
	a. 0001	b. 1101	c. 0101	d. 1001	
10	The register used to store the flags is called as _____				1
	a. Flag register	b. Status register	c. Test register	d. Log register	
11	The smallest entity of memory is called _____				1
	a. Cell	b. Block	c. Instance	d. Unit	
12	A _____ is used to restore the contents of the cells.				1
	a. Sense amplifier	b. Refresh counter	c. Restorer	d. None of the mentioned	
Q. 2	Solve the following.				12
A)	Explain the different components of Central Processing Unit?				6
B)	Explain in detail Instruction Fetching and Execution				6
Q.3	Solve the following.				12
A)	Explain different addressing modes.				6
B)	Explain RISC and CISC architecture.				6
Q. 4	Solve Any Two of the following.				12
A)	Briefly explain the following representations: sign magnitude, twos complement.				6
B)	Write a Booth algorithm for multiplication with a suitable example.				6
C)	Write a short note i) Magnetic Disk ii) Magnetic Tape				6
Q.5	Solve Any Two of the following.				12
A)	Explain Programmed I/O module and Interrupt driven I/O.				6
B)	Explain types of semiconductor memories technologies.				6
C)	Explain RAID memory in detail?				6
Q. 6	Solve Any Two of the following.				12
A)	Explain the Flynn's classification.				6
B)	Write short note on i) I/O channels ii) Direct memory Access (DMA)				6
C)	Explain the Micro programmed Control Unit?				6